# Exploratory Data Analysis of the Melanoma dataset

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**Dataset:** Survival from Malignant Melanoma (University Hospital of Odense, Denmark, 1962–1977)

**Software:** R (screenshots of code and figures are included from my RStudio output)

# 1. Introduction

This report explores the “Survival from Malignant Melanoma” dataset, which contains clinical and demographic measurements on 205 patients treated surgically at Odense University Hospital (Denmark) between 1962 and 1977, with follow up until the end of 1977 **(Arel-Bundock V., n.d**). The dataset includes the following variables:

- time: survival time in days since operation
- status: patient status at end of study (1 died from melanoma, 2 alive, 3 died from other causes)
- sex: 1 male, 0 female
- age: age at operation (years)
- year: year of operation
- thickness: tumour thickness (mm)
- ulcer: ulceration (1 present, 0 absent)

Thickness (often referred to clinically as Breslow thickness) and ulceration are widely recognised as prognostic features in melanoma. Thicker tumours and ulceration are generally associated with worse outcomes, although effect sizes and relationships can vary across subtypes and settings **(Bønnelykke-Behrndtz, 2017).**

# 2. Summary Statistics

## 2.1 Numerical Summary

```
> str(mel)
tibble [205 × 7] (S3: tbl_df/tbl/data.frame)
 $ time     : num [1:205] 10 30 35 99 185 204 210 232 232 279 ...
 $ status   : num [1:205] 3 3 2 3 1 1 1 3 1 1 ...
 $ sex      : num [1:205] 1 1 1 0 1 1 1 0 1 0 ...
 $ age      : num [1:205] 76 56 41 71 52 28 77 60 49 68 ...
 $ year     : num [1:205] 1972 1968 1977 1968 1965 ...
 $ thickness: num [1:205] 6.76 0.65 1.34 2.9 12.08 ...
 $ ulcer    : num [1:205] 1 0 0 0 1 1 1 1 1 1 ...
> summary(mel)
      time          status           sex              age             year        thickness          ulcer      
 Min.   :  10   Min.   :1.00   Min.   :0.0000   Min.   : 4.00   Min.   :1962   Min.   : 0.10   Min.   :0.000  
 1st Qu.:1525   1st Qu.:1.00   1st Qu.:0.0000   1st Qu.:42.00   1st Qu.:1968   1st Qu.: 0.97   1st Qu.:0.000  
 Median :2005   Median :2.00   Median :0.0000   Median :54.00   Median :1970   Median : 1.94   Median :0.000  
 Mean   :2153   Mean   :1.79   Mean   :0.3854   Mean   :52.46   Mean   :1970   Mean   : 2.92   Mean   :0.439  
 3rd Qu.:3042   3rd Qu.:2.00   3rd Qu.:1.0000   3rd Qu.:65.00   3rd Qu.:1972   3rd Qu.: 3.56   3rd Qu.:1.000  
 Max.   :5565   Max.   :3.00   Max.   :1.0000   Max.   :95.00   Max.   :1977   Max.   :17.42   Max.   :1.000  
```

### Commentary

- Survival time shows wide variability with a large range. The mean is slightly above the median, suggesting some right skew (longer survivors pulling up the mean). The minimum of 10 days indicates at least one very short survival after surgery, which is likely to appear as an outlier on boxplots.

- Age is broadly spread across adulthood and older ages, with one very young observation (4 years) that will appear as an outlier. Mean and median are close, suggesting age is not extremely skewed.
- Operations cluster around 1970, with most data in the late 1960s to early 1970s. The tight spread is expected due to the study period.
- Thickness is strongly right skewed. The mean is higher than the median and the maximum (17.42 mm) is far above the upper quartile (3.56 mm). This shape often produces non normality in QQ plots and motivates nonparametric tests as a robustness check.

## 2.2 Categorical Summary

```
> table(mel$status)

  1   2   3 
 57 134  14 
> prop.table(table(mel$status))

         1          2          3 
0.27804878 0.65365854 0.06829268 
> table(mel$sex)

  0   1 
126  79 
> prop.table(table(mel$sex))

        0         1 
0.6146341 0.3853659 
> table(mel$ulcer)

  0   1 
115  90 
> prop.table(table(mel$ulcer))

        0         1 
0.5609756 0.4390244 
```

### Sex

- Female: **126** (61.46%)
- Male: **79** (38.54%)

### Status

- Alive: **134** (65.37%)
- Died from melanoma: **57** (27.80%)
- Died from other causes: **14** (6.83%)

### Ulceration

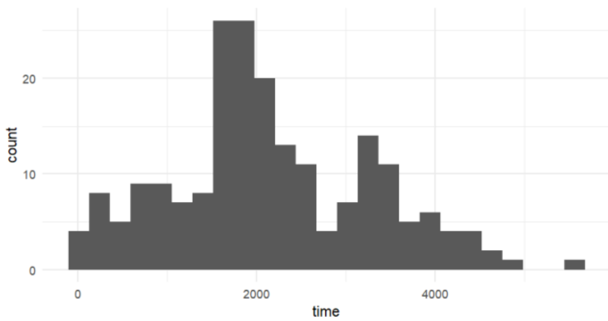
- Absent: **115** (56.10%)
- Present: **90** (43.90%)

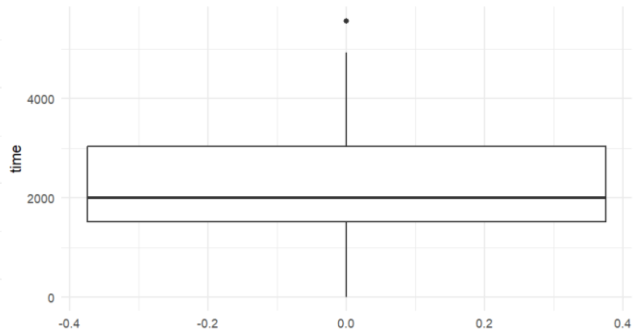
### Commentary

The dataset contains more females than males. Most patients were alive at the end of follow up (about two thirds), with melanoma related deaths forming the largest death category. Ulceration is present in a substantial minority (about 44%), which is clinically important because ulceration is frequently treated as an adverse prognostic indicator.

# 3. Graphical Summaries

### 3.1 Time histogram and boxplot

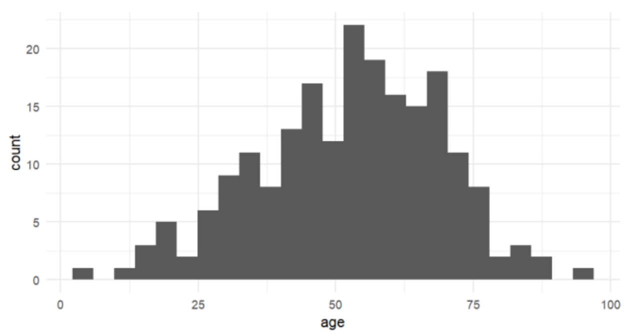


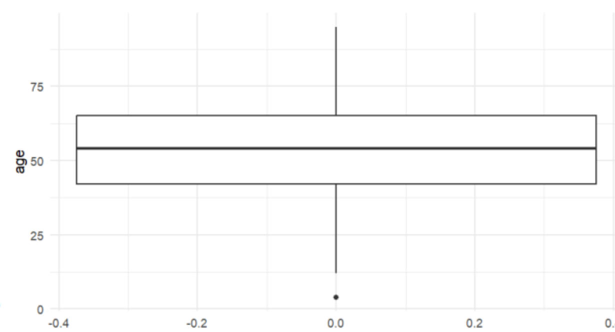




**Interpretation**

Survival time is not tightly concentrated, and the presence of outliers suggests that purely normal based assumptions may be questionable in some subgroups.

### 3.2 Age histogram and boxplot

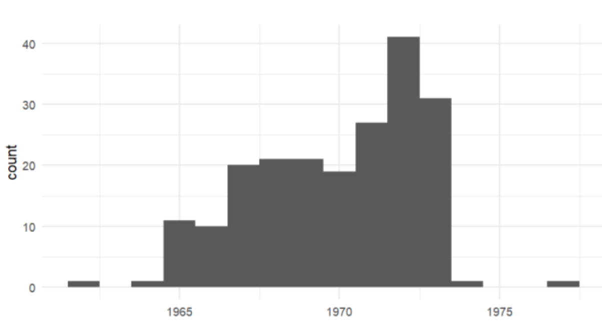



**Interpretation**

- Age often looks approximately symmetric with mild tails
- Boxplot highlights the young outlier (age 4)

### 3.3 Year histogram and boxplot



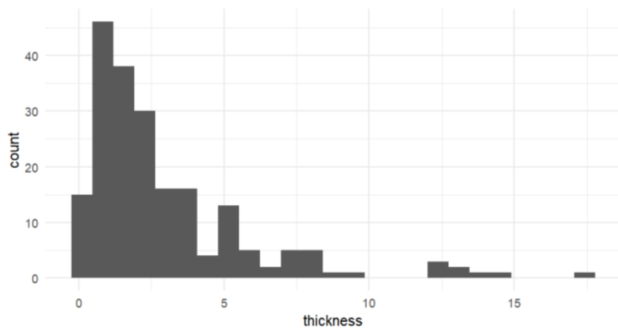


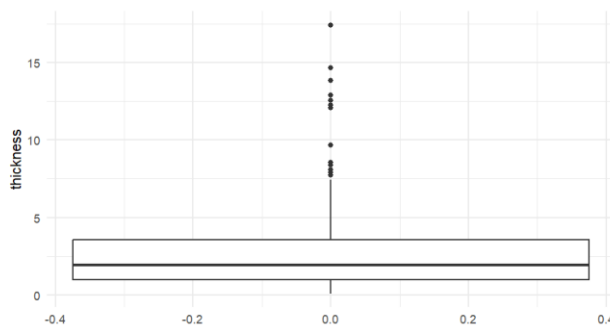
**Pattern**

- Discrete values spanning around 1962 and 1977
- Concentration around 1968 to 1972

**Interpretation**
This is a study design variable. It may be useful later to check whether treatment or follow up differences across years could confound outcomes.

### 3.4 Thickness histogram and boxplot



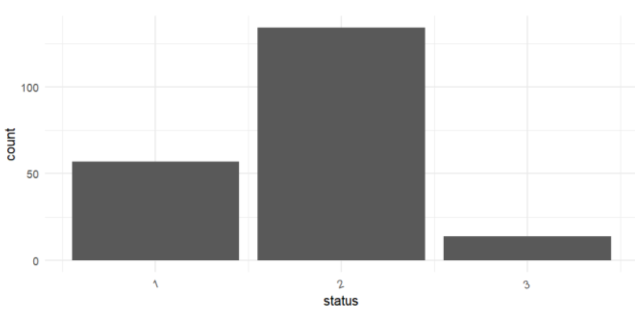

**Pattern**

- Strong right skew with a long upper tail
- Several high thickness outliers

**Interpretation**
Thickness clearly violates normality visually. This matters when choosing tests and interpreting linear regression residuals.

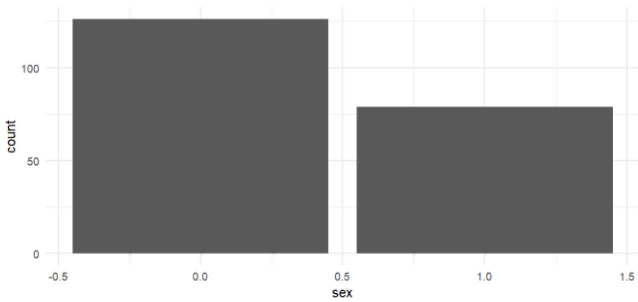
### 3.5 Graphs for Categorical variables

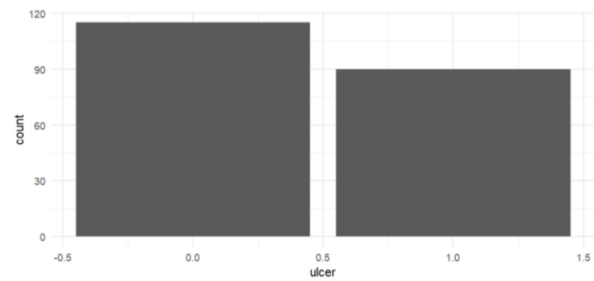
### 3.5.1 Bar chart for Status




**Interpretation**
Most patients are alive, even though a substantial fraction died from melanoma. The outcome is not purely binary because deaths from other causes exist. This is relevant if you later extend the work to proper survival analysis.

### 3.5.2 Bar Charts for sex and ulcer



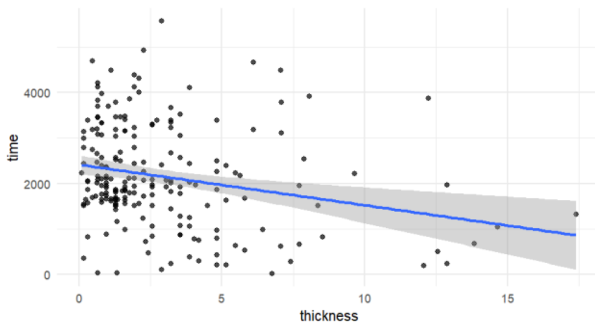




**Interpretation**

The sample is female majority. Ulceration is common enough that it could be examined in deeper modelling.

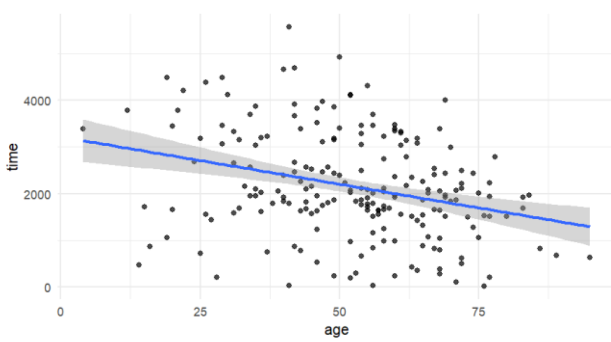
# 4. Regression Analysis and correlation

### 4.1 Time and Thickness



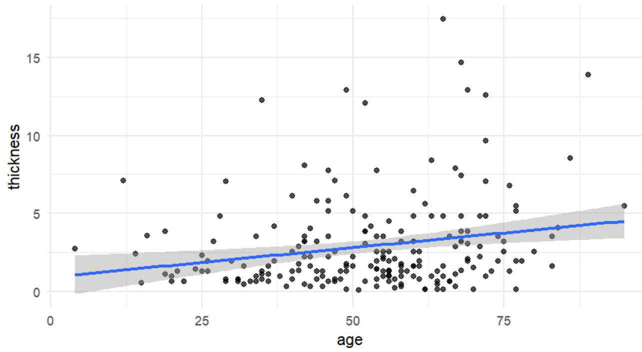

```
> summary(m1)

Call:
lm(formula = time ~ thickness, data = mel)

Residuals:
    Min      1Q  Median      3Q     Max 
-2325.4  -707.6  -210.6   744.9  3410.4 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  2413.41     107.39  22.473  < 2e-16 ***
thickness     -89.25      25.86  -3.451 0.000679 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1093 on 203 degrees of freedom
Multiple R-squared:  0.05542,	Adjusted R-squared:  0.05076 
F-statistic: 11.91 on 1 and 203 DF,  p-value: 0.0006793
```

### 4.2 Time and age




```
> summary(m2)

Call:
lm(formula = time ~ age, data = mel)

Residuals:
    Min      1Q  Median      3Q     Max 
-2464.3  -646.2   -54.4   712.1  3179.6 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 3217.448    247.879  12.980  < 2e-16 ***
age          -20.293      4.504  -4.506 1.12e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1072 on 203 degrees of freedom
Multiple R-squared:  0.09091,	Adjusted R-squared:  0.08643 
F-statistic:  20.3 on 1 and 203 DF,  p-value: 1.116e-05
```

### 4.3 Thickness and age




```
> summary(m3)

Call:
lm(formula = thickness ~ age, data = mel)

Residuals:
    Min      1Q  Median      3Q     Max 
-3.6853 -1.7727 -0.9155  0.9558 14.0273 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)   
(Intercept)  0.94105    0.67004   1.404  0.16170   
age          0.03772    0.01217   3.098  0.00222 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.899 on 203 degrees of freedom
Multiple R-squared:  0.04515,	Adjusted R-squared:  0.04044 
F-statistic: 9.598 on 1 and 203 DF,  p-value: 0.002223
```

## 5. Commentary on observed relationships

### 5.1 Thickness and survival time

The negative slope for time ~ thickness indicates that thicker tumours tend to be associated with shorter survival times. Quantitatively, the model suggests an average decrease of about 89 days of survival per additional 1 mm of thickness. This aligns with clinical expectations that greater tumour depth is associated with poorer prognosis (Breslow thickness is widely used as a staging and prognostic indicator) (**El Sharouni et al., 2022**).

However, the model $R^2$is only 0.055, meaning thickness alone explains about 5.5% of the variation in survival time. This is not a weakness of the analysis. It reflects that survival is influenced by many factors not included here (ulceration, stage, nodal involvement, treatment differences, competing causes of death, and unmeasured patient factors).

### 5.2 Age and survival time

The relationship between age and survival is also negative and statistically strong. The estimate suggests around 20 fewer days of survival per additional year of age. Again, explanatory power is modest ($R^2 = 0.091$), implying age is associated with survival but does not determine it alone.

### 5.3 Age and thickness

Thickness increases slightly with age. The slope implies that a 10-year increase in age corresponds to around 0.38 mm higher thickness on average. While statistically significant, $R^2 = 0.045$ indicates substantial unexplained variation. A plausible interpretation is that older patients may present with more advanced lesions or there may be biological or health system factors influencing detection timing.

### 5.4 Limitations of linear regression in this context

- time is a survival outcome and is typically handled with survival methods that properly account for censoring and competing risks. The current regression treats time as a continuous response without modelling censoring formally.

- Thickness is strongly skewed, which can influence residual behaviour and the linear fit. A log transform of thickness would be a reasonable extension, though not required by the brief.
- Relationships may be non-linear (for example, thickness effects may not be constant across its full range).

## 6. Two sample significance tests grouped by gender

Means from our dataset:

- **Survival time** mean: Female 2282.64, Male 1945.71
- **Thickness** mean: Female 2.49, Male 3.61
- **Age** mean: Female 51.56, Male 53.90

### Hypotheses and test choices

For each variable $X \in \{time, age, thickness\}$:

- $H_0$: mean of $X$ is equal for males and females
- $H_1$: mean of $X$ differs by gender

**Primary test:** Welch two sample t-test (robust to unequal variances). **Robustness check:** Wilcoxon rank sum test due to skewness (especially for thickness and potentially time).

```
+   summarise(
+     n = n(),
+     time_mean = mean(time), time_sd = sd(time), time_median = median(time),
+     age_mean = mean(age), age_sd = sd(age), age_median = median(age),
+     thick_mean = mean(thickness), thick_sd = sd(thickness), thick_median = median(thickness)
+   )
# A tibble: 2 × 11
    sex     n time_mean time_sd time_median age_mean age_sd age_median
  <dbl> <int>     <dbl>   <dbl>       <dbl>    <dbl>  <dbl>      <dbl>
1     0   126     2283.   1090.        2059     51.6   16.1         54
2     1    79     1946.   1148.        1860     53.9   17.6         55
# ℹ 3 more variables: thick_mean <dbl>, thick_sd <dbl>,
#   thick_median <dbl>
> t.test(time ~ sex, data = mel)

	Welch Two Sample t-test

data:  time by sex
t = 2.0848, df = 159.27, p-value = 0.03868
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
95 percent confidence interval:
  17.74767 656.12032
sample estimates:
mean in group 0 mean in group 1 
       2282.643        1945.709 
```

```
> t.test(age ~ sex, data = mel)

	Welch Two Sample t-test

data:  age by sex
t = -0.95559, df = 154.42, p-value = 0.3408
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
95 percent confidence interval:
 -7.162764  2.492280
sample estimates:
mean in group 0 mean in group 1 
       51.56349        53.89873 

> t.test(thickness ~ sex, data = mel)

	Welch Two Sample t-test

data:  thickness by sex
t = -2.6059, df = 149.09, p-value = 0.01009
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
95 percent confidence interval:
 -1.9775560 -0.2718653
sample estimates:
mean in group 0 mean in group 1 
       2.486429        3.611139 
```

```
> wilcox.test(time ~ sex, data = mel)

	Wilcoxon rank sum test with continuity correction

data:  time by sex
W = 5824.5, p-value = 0.04046
alternative hypothesis: true location shift is not equal to 0

> wilcox.test(age ~ sex, data = mel)

	Wilcoxon rank sum test with continuity correction

data:  age by sex
W = 4587.5, p-value = 0.3466
alternative hypothesis: true location shift is not equal to 0

> wilcox.test(thickness ~ sex, data = mel)

	Wilcoxon rank sum test with continuity correction

data:  thickness by sex
W = 3794.5, p-value = 0.004213
alternative hypothesis: true location shift is not equal to 0
```

### 6.1 Results

#### 6.1.1 Survival time by gender

- Welch t test p = 0.0387
- Wilcoxon p = 0.0405

**Conclusion:** Evidence at the 5% level that survival time differs by gender, with females showing higher average survival time.

**6.1.2 Age by gender**

- Welch t test p = 0.3408
- Wilcoxon p = 0.3466

**Conclusion:** No evidence of a gender difference in age distribution in this sample.

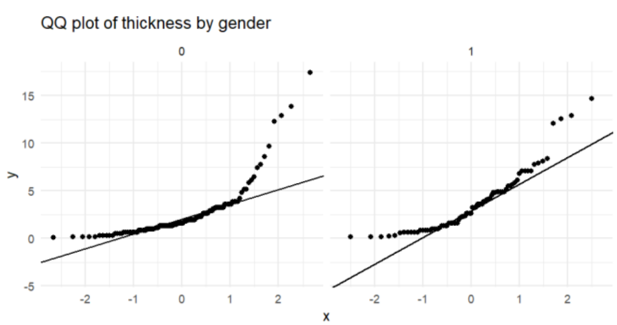
**6.1.3 Thickness by gender**

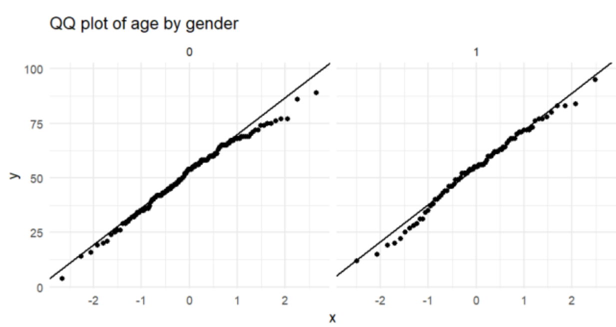
- Welch t test p = 0.0101
- Wilcoxon p = 0.00421

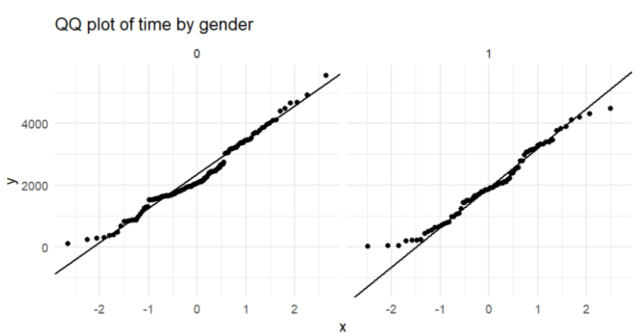
**Conclusion:** Strong evidence that thickness differs by gender, with males having higher thickness on average.

**Interpretation:** The observed higher thickness in males could contribute to worse outcomes if thickness is prognostically adverse. This is consistent with broader clinical framing where tumour thickness and ulceration are key indicators of risk **(Eguchi, M.M. et al., 2023).**

# 7. QQ-plots




**Commentary**

**Thickness:** Thickness shows major deviation from the straight line in both genders, especially in the upper tail, indicating strong right skew and non-normality. This supports using the Wilcoxon test as a robustness check and suggests that transformations (such as log thickness) could be explored in extended modelling.

**Age:** Age is comparatively close to normal in both genders, with only mild tail deviations. This supports the appropriateness of t-tests for age.

**Survival time:** Time typically shows mild to moderate departures from normality (often tail deviations). This is expected for survival type outcomes. The QQ plots support cautious interpretation of normal theory methods and again justify reporting both Welch and Wilcoxon tests.

## 8. Insights and Recommendations

### 8.1 Key insights from the EDA

1. **Thickness is negatively associated with survival time.**
   The relationship is statistically significant and directionally consistent with clinical understanding that thicker tumours are associated with worse prognosis (**El Sharouni et al., 2022**).

2. **Age is also negatively associated with survival time.**
   Older patients show lower survival time on average, though the relationship still explains only a modest proportion of variability.

3. **Gender differences appear in thickness and survival time.**
   Males have significantly greater tumour thickness, and females show higher survival time on average. Age does not differ significantly by gender in this sample, suggesting age is unlikely to explain gender differences here.

4. **Thickness is strongly non normal.**
   This is visible in histograms and QQ plots and is supported by the consistency of non-parametric tests. This distributional behaviour is important when selecting models and interpreting regression assumptions.

5. **Ulceration is common and likely important but not used in the required regressions.**
   Given ulceration's known prognostic relevance, it is a strong candidate for deeper modelling beyond the required scope **(Bønnelykke-Behrndtz, 2017).**

### 8.2 Recommendations for further investigation

1. **Extend to survival analysis methods (recommended next step).**
   Because time is a survival outcome and status captures different end states; a Cox proportional hazards model (or competing risks approach) would be more appropriate than

linear regression for causal style interpretation. The R survival package is designed for this **(Therneau, T.M., 2024)**

2. **Include ulceration explicitly.**
   A multivariable model examining thickness and ulceration together is recommended, because both are clinically meaningful and may interact. Ulceration is commonly described as prognostically adverse **(Bønnelykke-Behrndtz, 2017).**

3. **Consider transformations for thickness.**
   Given the right skew, reanalysing with log thickness could stabilise variance and improve model fit.

4. **Check nonlinearity and interactions.**
   Scatterplots may suggest that effects are not perfectly linear across the full thickness range. Models with splines or stratification (thin vs thick) could be explored.

5. **Investigate whether year of operation affects outcomes.**
   Although the time window is limited, clinical practice may have evolved over time. Checking whether year correlates with thickness or survival could detect cohort effects.

## 9. Conclusion

This exploratory analysis of 205 melanoma patients found statistically significant associations consistent with clinical expectations: **greater tumour thickness and older age are linked to shorter survival time**, and **males show higher tumour thickness than females**. Thickness is strongly non normal, which was addressed by using both parametric and non-parametric comparisons for gender-based tests. The findings motivate deeper modelling using survival methods and inclusion of ulceration, which is widely regarded as prognostically important.

## References


Arel-Bundock, V. (n.d.) *Rdatasets: Survival from Malignant Melanoma (Melanoma_df)*.

Bønnelykke-Behrndtz, M.L. (2017) *Ulcerated Melanoma: Aspects and Prognostic Impact*. NCBI Bookshelf.

Eguchi, M.M. et al. (2023) ‘Prognostic modeling of cutaneous melanoma stage I patients…’, *Cancer* (Wiley).

El Sharouni, M.A. et al. (2022) ‘The progressive relationship between increasing Breslow thickness and decreasing survival…’, *Journal of the American Academy of Dermatology*.

Therneau, T.M. (2024) *A package for survival analysis in R* (vignette). CRAN.